CVE 255: Basics of Drone Technology

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Academic inclusion - Necessity

- Placements and career advancements
- Emerging technologies
- Student research
- Technology adoption
- Increasing drone - job market



Learning Objectives

1. **Objective 1:** To understand the historical evolution and various classifications.
2. **Objective 2:** To learn the fundamental principles of aerodynamics and flight mechanics in UAVs.
3. **Objective 3:** To introduce the key propulsion systems and structural materials used in drones.
4. **Objective 4:** To develop an understanding of India's drone regulatory framework (DGCA rules).



Unmanned Aerial Vehicles (UAV)

UAVs are aircrafts that do not have a human pilot onboard but can be operated at varying levels of autonomy.



Architecture classification: Fixed wing, Rotary wing, Hybrid

Size classification: Micro, Small, Medium, Large

Altitude classification: low range, Medium range, High range

Payloads: Cameras, Sensors, Microphones



Terminology

UAV Acronyms

- Upper Atmosphere Vehicle
- Unmanned Autonomous Vehicle
- Unmanned Airborne Vehicle
- Uninhabited Aircraft Vehicle
- Unmanned Aerospace Vehicle
- Unmanned Aircraft Vehicle
- Unmanned Air Vehicle

Differentiation

UAS : Unmanned Aerial System

UAV : Unmanned Aerial Vehicle

Drone : Dynamic Remotely Operated Navigation Equipment

RPAS : Remotely Piloted Aerial Systems

RPA : Remotely Piloted Aircraft

RPV : Remotely piloted vehicle

Drones

- Drones are a class of unmanned aerial vehicles, which can take to the air without human pilots.
- Drones have software-controlled flight plans integrated into their systems
- These systems work with GPS and a controller to guide and track their movements.



Quadcopter



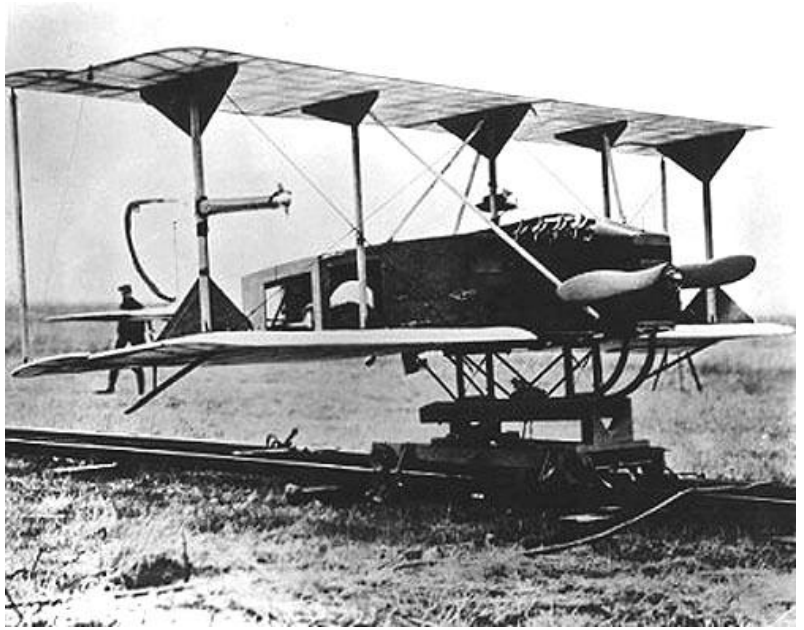
Fixed Wing Rotor



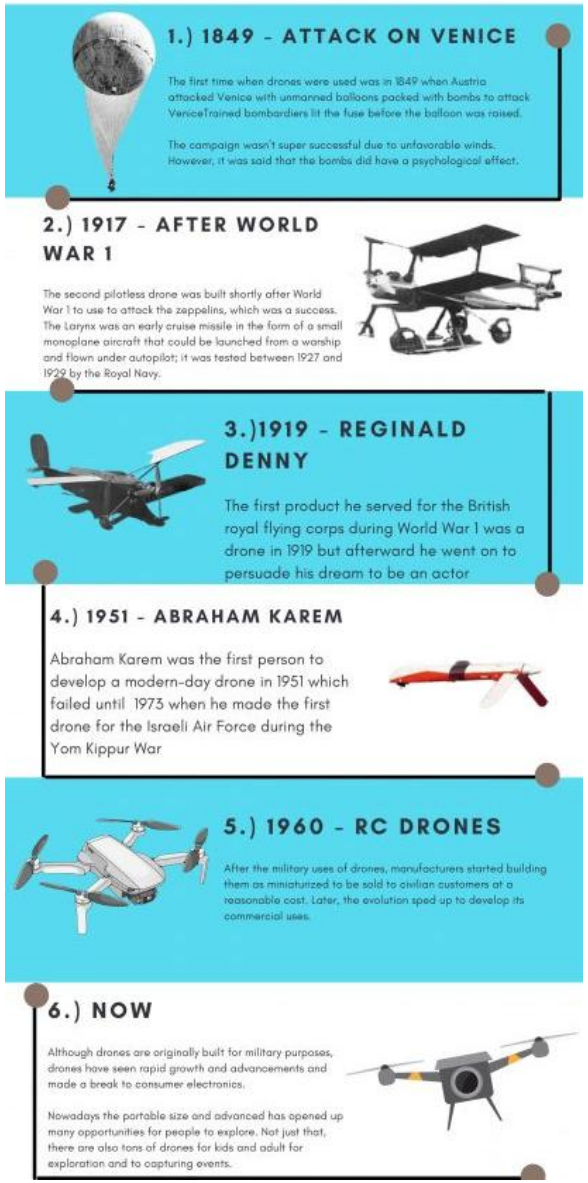
Hybrid Drones

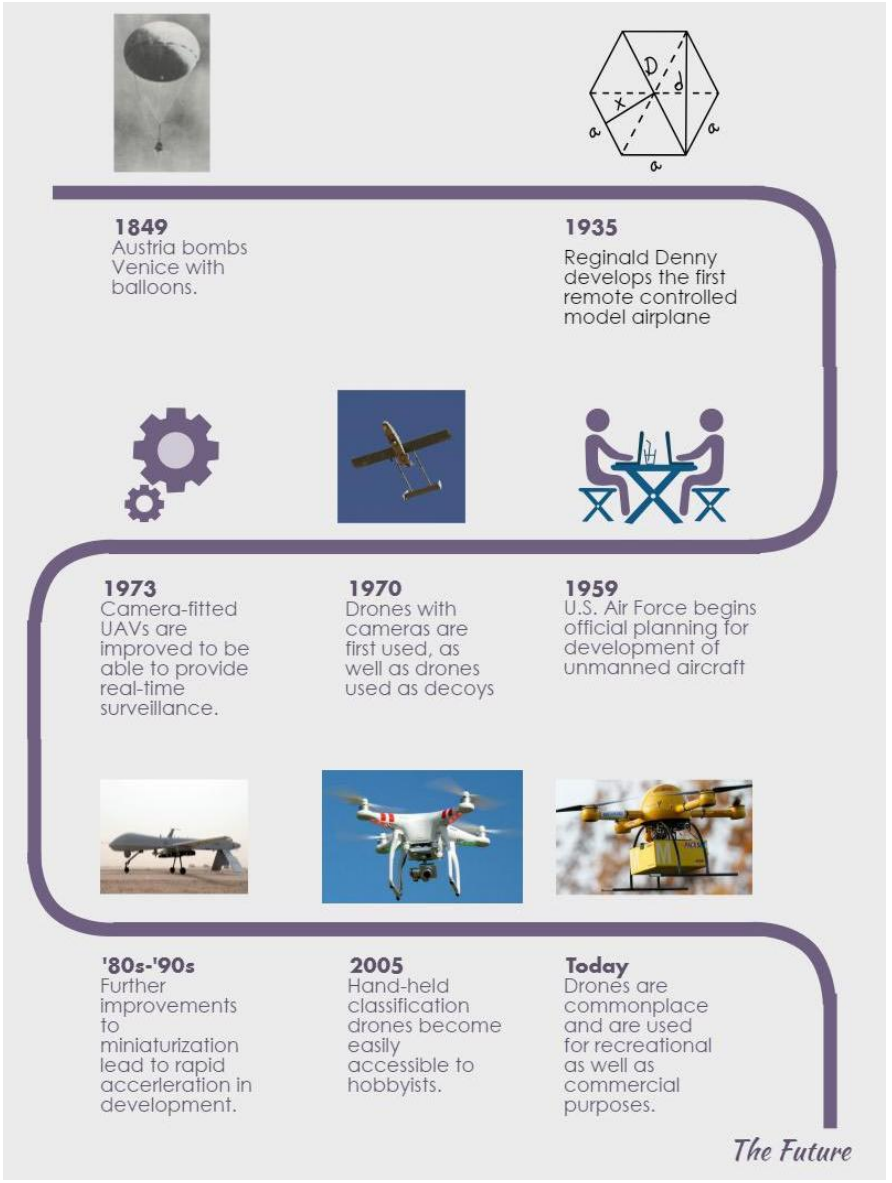
Origin of Drones

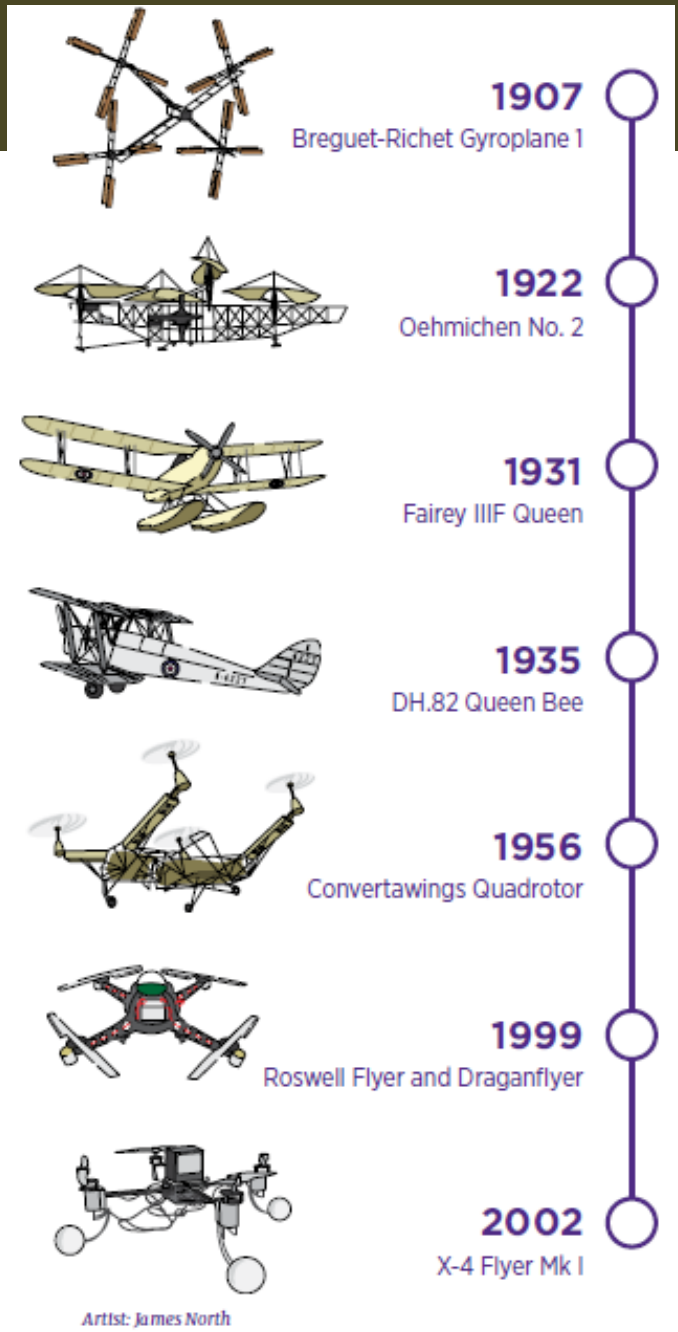
- The Hewitt-Sperry Automatic Airplane was a project undertaken during World War I to develop an aerial torpedo, also called a flying bomb or pilotless aircraft
- Sperry had been perfecting gyroscopes for naval use since 1896 and established the Sperry Gyroscope Company in 1910.
- First flight : September 1917
- Abraham Karem – Father of Drone Technology
  - Founded Leading Systems Inc. in his home garage
  - First drone, Albatross, and later Amber, which eventually evolved into the famous General Atomics MQ-1 Predator drone.



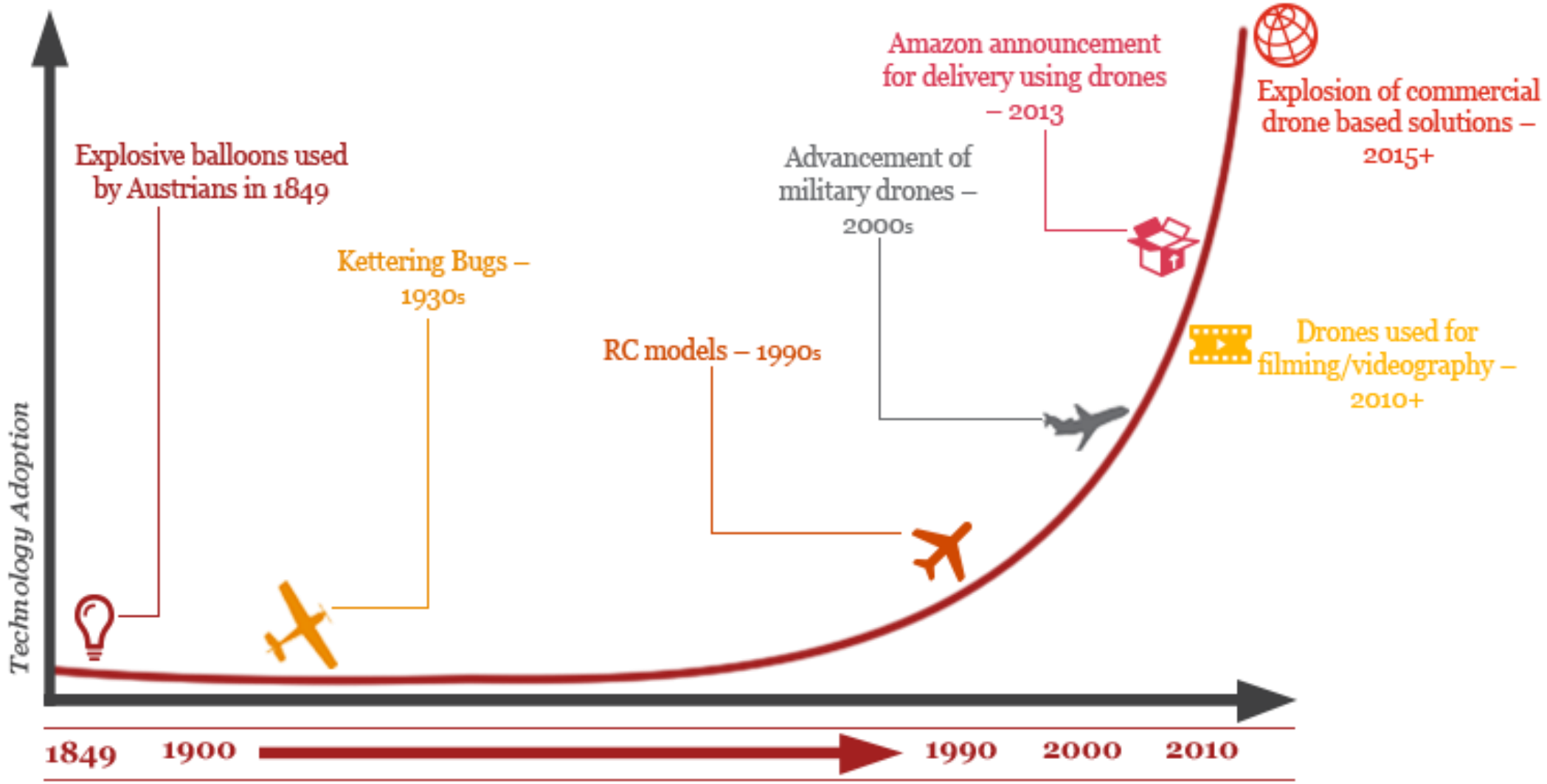
Evolution


1.) 1849 - ATTACK ON VENICE
The first time when drones were used was in 1849 when Austria attacked Venice with unmanned balloons packed with bombs to attack VeniceTrained bombardiers lit the fuse before the balloon was raised.
The campaign wasn't super successful due to unfavorable winds. However, it was said that the bombs did have a psychological effect.
2.) 1917 - AFTER WORLD WAR 1
The second pilotless drone was built shortly after World War 1 to use to attack the zeppelins, which was a success. The Larynx was an early cruise missile in the form of a small monoplane aircraft that could be launched from a warship and flown under autopilot; it was tested between 1927 and 1929 by the Royal Navy.
3.)1919 - REGINALD DENNY
The first product he served for the British royal flying corps during World War 1 was a drone in 1919 but afterward he went on to persuade his dream to be an actor
4.) 1951 - ABRAHAM KAREM
Abraham Karem was the first person to develop a modern-day drone in 1951 which failed until 1973 when he made the first drone for the Israeli Air Force during the Yom Kippur War
5.) 1960 - RC DRONES
After the military uses of drones, manufacturers started building them as miniaturized to be sold to civilian customers at a reasonable cost. Later, the evolution sped up to develop its commercial uses.
6.) NOW
Although drones are originally built for military purposes, drones have seen rapid growth and advancements and made a break to consumer electronics.
Nowadays the portable size and advanced has opened up many opportunities for people to explore. Not just that, there are also tons of drones for kids and adult for exploration and to capturing events.


1849
Austria bombs Venice with balloons.
1935
Reginald Denny develops the first remote controlled model airplane
1973
Camera-fitted UAVs are improved to be able to provide real-time surveillance.
1970
Drones with cameras are first used, as well as drones used as decoys
1959
U.S. Air Force begins official planning for development of unmanned aircraft
'80s-'90s
Further improvements to miniaturization lead to rapid accerleration in development.
2005
Hand-held classification drones become easily accessible to hobbyists.
Today
Drones are commonplace and are used for recreational as well as commercial purposes.
The Future

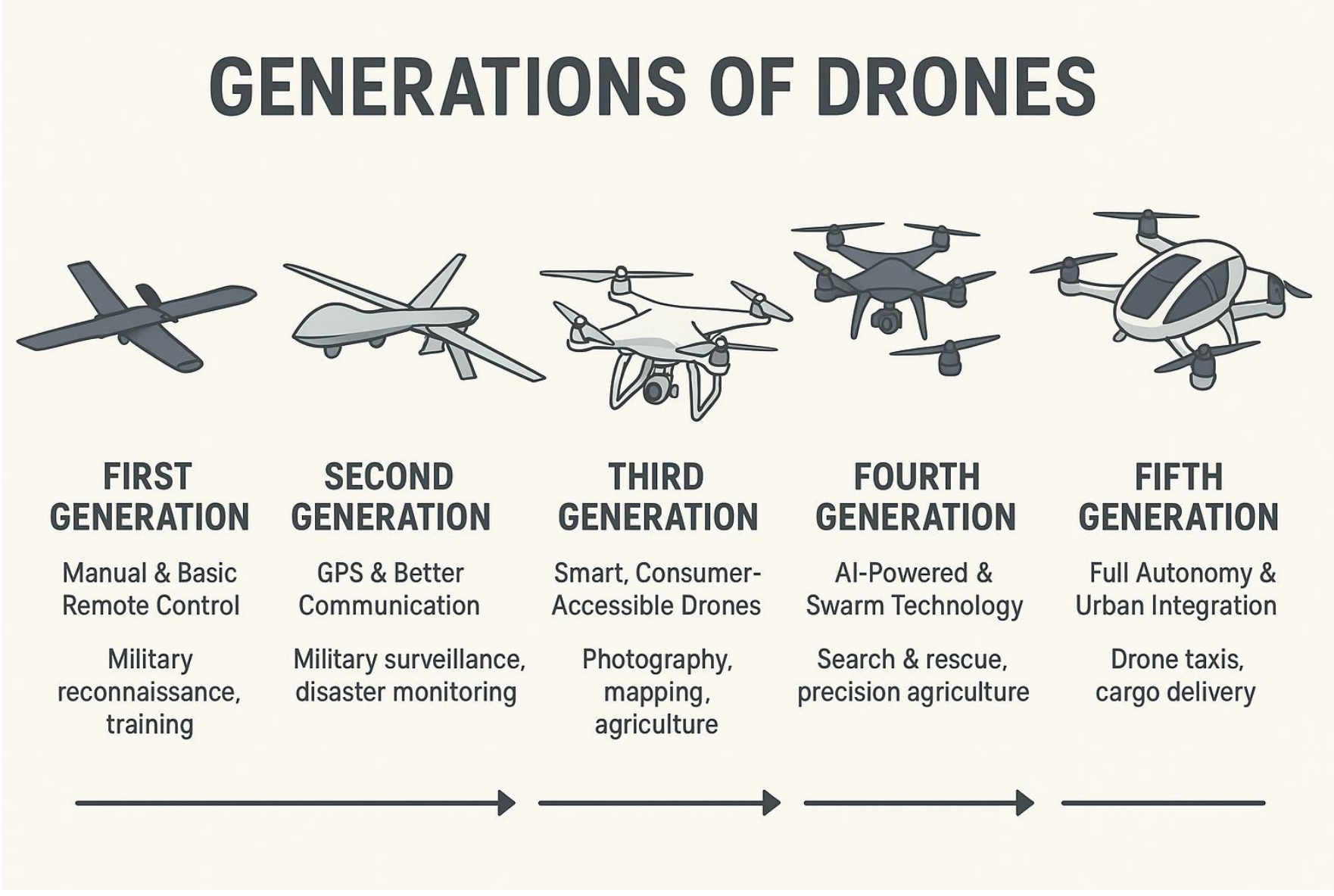

1907
Breguet-Richet Gyroplane 1
1922
Oehmichen No. 2
1931
Fairey IIIF Queen
1935
DH.82 Queen Bee
1956
Convertawings Quadrotor
1999
Roswell Flyer and Draganflyer
2002
X-4 Flyer Mk I
Artist: James North

Evolution

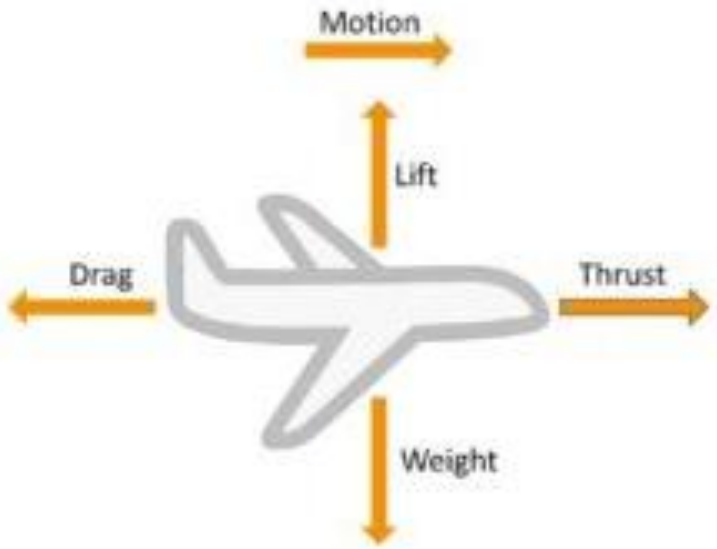

Explosive balloons used by Austrians in 1849
Kettering Bugs – 1930s
RC models – 1990s
Advancement of military drones – 2000s
Amazon announcement for delivery using drones – 2013
Explosion of commercial drone based solutions – 2015+
Drones used for filming/videography – 2010+
Technology Adoption
1849
1900
1990
2000
2010

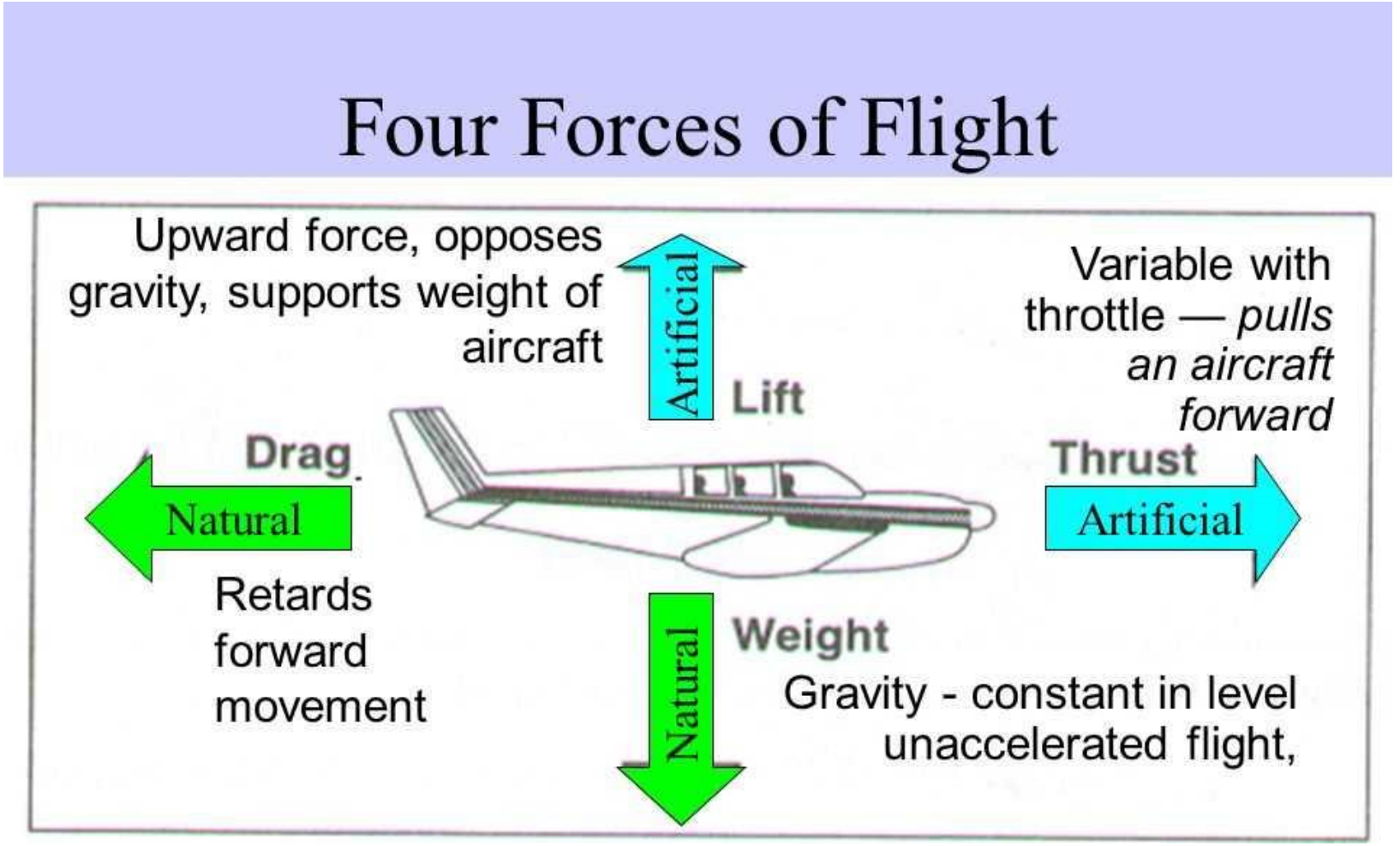
Generations of Drones

- *First & Second Gen*: Defense and reconnaissance.
- *Third Gen*: Commercial photography, surveying.
- *Fourth Gen*: Infrastructure inspection, smart agriculture.
- *Fifth Gen*: Passenger transport, autonomous logistics.



Forces in aerial systems



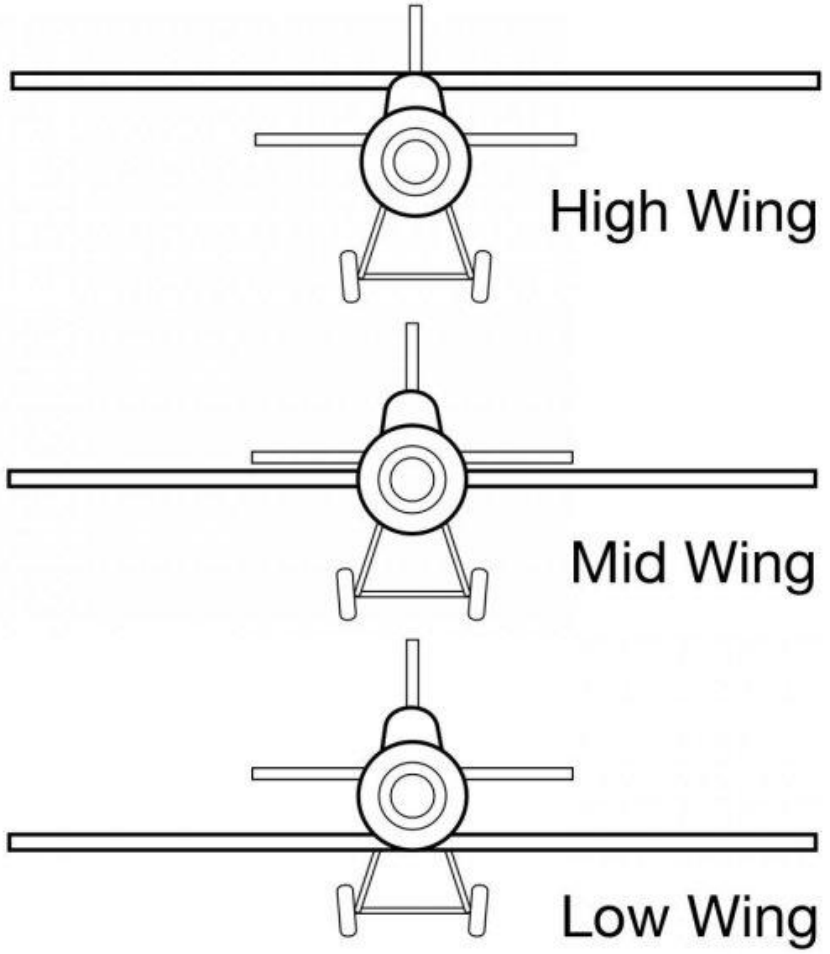


Fixed Wing Drones

- Fixed-Wing Drones Characteristics: Resemble traditional airplanes. Wings provide lift instead of vertical thrust.

- Advantages: Longer flight times.
- Greater range.
- Disadvantages: Require runways or catapult
- Cannot hover.

- Applications: Mapping and surveying.
- Agricultural monitoring.
- Long-range surveillance.



Rotary Wing Drones

- Types:
- Single-Rotor : Similar to helicopters.
- Multi-Rotor : Includes tricopters, quadcopters, hexacopters, and octocopters.



- Characteristics : Vertical takeoff and landing.
- Ability to hover.
- Advantages: Stability and maneuverability.
- Easier to use in confined spaces.
- Disadvantages: Shorter flight times compared to fixed-wing.
- Limited range.
- Applications: Aerial photography and videography.
- Inspection and maintenance.
- Law enforcement and search and rescue.



Single Rotor Drones

- Characteristics: One main rotor and a tail rotor.
- Similar to traditional helicopters.

- Advantages: Higher efficiency and longer flight times than multi-rotors.
- Ability to carry heavier payloads.



- Disadvantages: More complex mechanics.
- Increased risk of damage.

- Applications: Aerial LiDAR (Light Detection and Ranging)
- Power line inspections.
- Heavy-lift operations.

Hybrid Drones

- Characteristics: Combine features of fixed-wing and rotary-wing drones.
- Vertical takeoff and landing (VTOL) with efficient forward flight.

- Advantages: Versatility in operations.
- Longer flight times than rotary-wing.

- Disadvantages: More complex and expensive.

- Applications: Delivery services.
  Environmental monitoring.
  Large-scale agricultural monitoring.



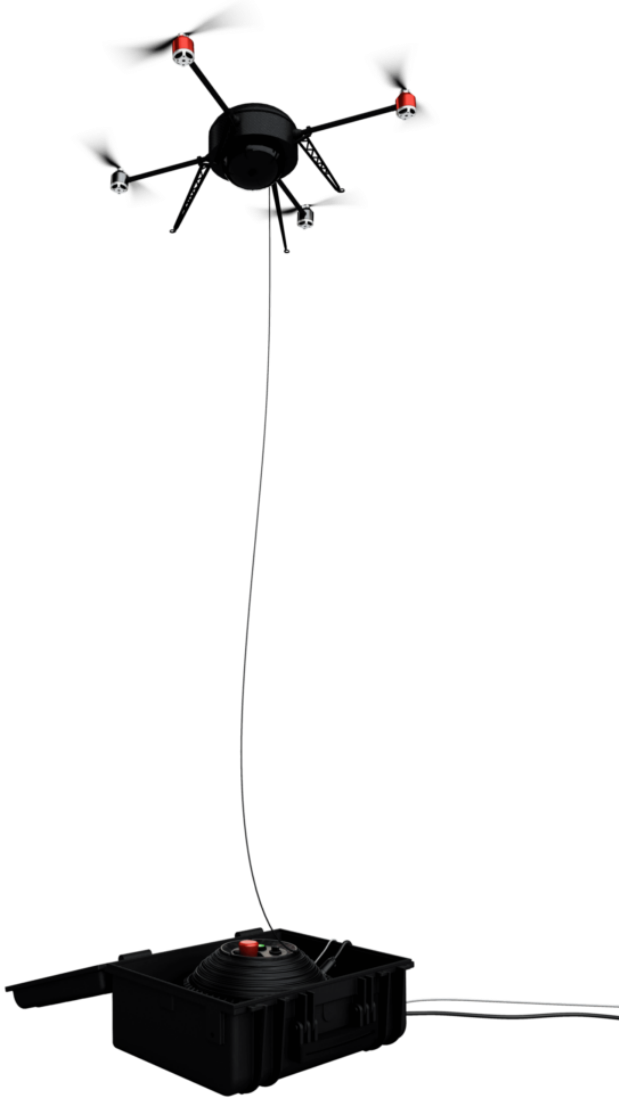
Tethered Drones

- Characteristics: Connected to a ground station via a tether.
- Provides continuous power and data transfer.

- Advantages: Unlimited flight time.
- Secure and stable operation.

- Disadvantages: Limited mobility due to tether.

- Applications: Surveillance and security.
- Live event broadcasting.
- Emergency response.



Micro and Nano Drones

- Characteristics: Very small and lightweight.

- Advantages: Highly portable.
- Can access confined spaces.

- Disadvantages: Limited flight time and range.
- Small payload capacity.

- Applications: Indoor inspections.
- Covert operations.
- Research and development.



Coaxial Drone

- Coaxial rotors can be employed, in which each arm has two motors, running in opposite directions (one facing up and one facing down).



Control Surfaces

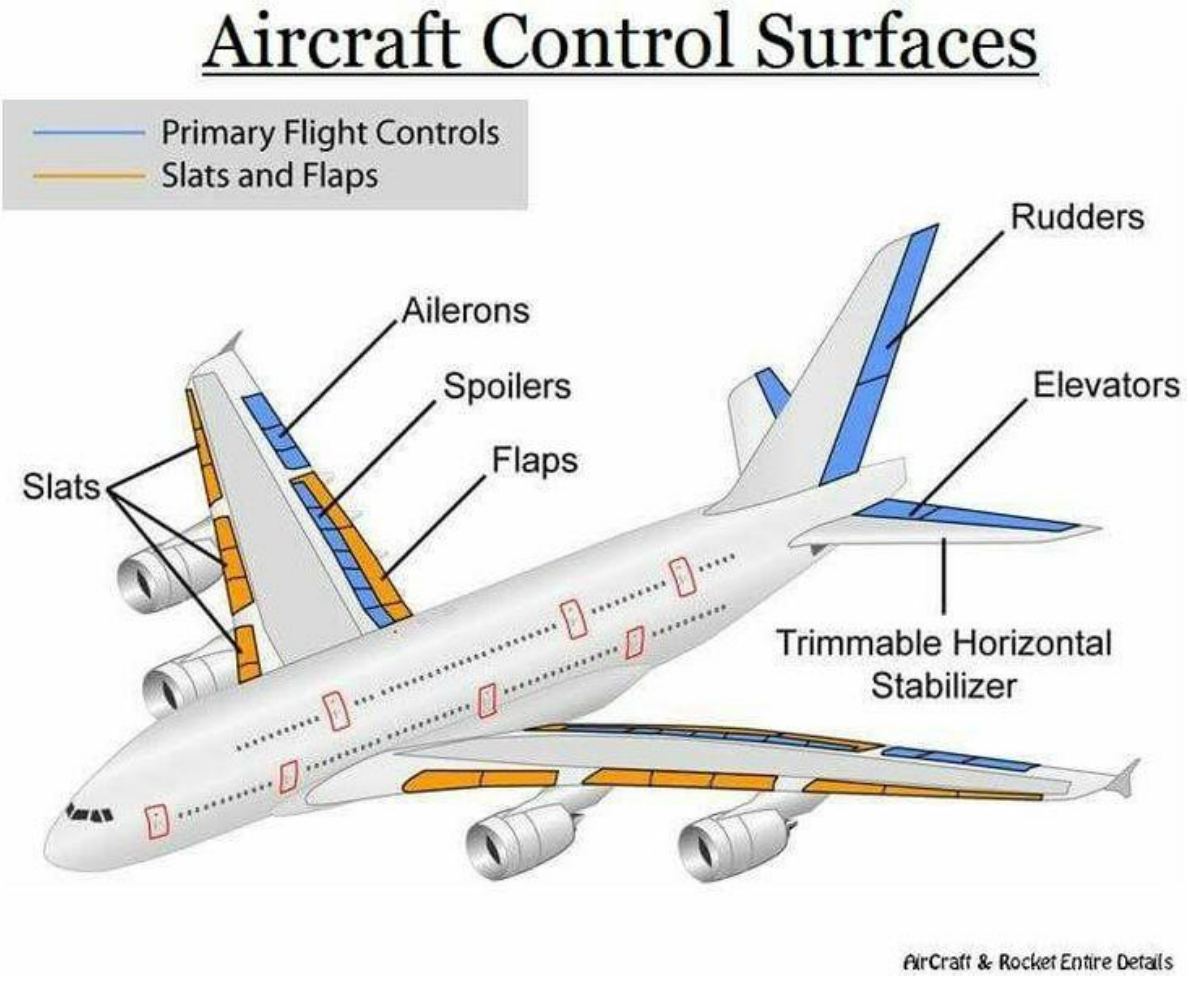
•Ailerons: Located on the wings, they control roll and are crucial for turning the aircraft.

•Elevators: Situated on the tail, they regulate pitch, allowing the aircraft to climb or descend.

•Rudder: Also on the tail, it manages yaw, facilitating left or right turns.

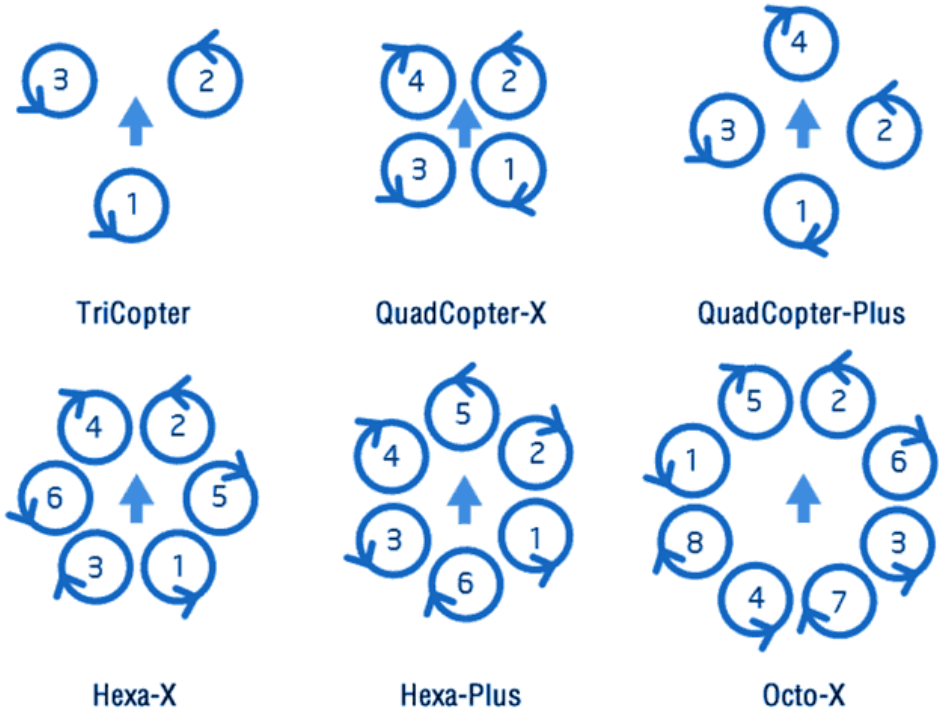
•Flaps: These are used to increase lift during takeoff and landing by changing the wing's shape.

•Slats: Located on the leading edge of the wings, they help improve airflow over the wing at high angles of attack. These surfaces work by creating differences in air pressure, allowing pilots to control the aircraft's attitude and orientation effectively.



Multirotor configurations

- A multirotor or multi copter is a rotorcraft with more than two rotors.
- Namely tricopter, quadcopter, hexacopter and octocopter, frequently used to refer to 3-, 4-, 6- and 8-rotor rotorcraft, respectively



Numericals

A drone of mass $m = 1.90\ \text{kg}$ produces a total thrust $T = 25.0\ \text{N}$ directed along the body axis pitched upward at $\theta = 20^\circ$ (i.e., thrust has forward and upward components). Find the horizontal acceleration a_x and vertical acceleration a_y.

Resolve components: $T_x = T\sin\theta, \quad T_y = T\cos\theta.$

Compute $\sin 20^\circ \approx 0.3420201,\ \cos 20^\circ \approx 0.9396926.$

$$T_y = 25.0 \times 0.9396926 \approx 23.4923\ \text{N}.$$
$$T_x = 25.0 \times 0.3420201 \approx 8.5505\ \text{N}.$$

Weight $mg = 1.90 \times 9.81 = 18.639\ \text{N}.$

Net vertical force $F_{y,\text{net}} = T_y - mg = 23.4923 - 18.639 = 4.8533\ \text{N}.$

Vertical accel $a_y = 4.8533/1.90 \approx 2.554\ \text{m/s}^2.$

Horizontal accel $a_x = T_x/m = 8.5505/1.90 \approx 4.5003\ \text{m/s}^2.$

Numericals

Mass $m = 2.00\ \text{kg}$. Drone tilts $\theta = 15^\circ$ and increases thrust to hold altitude. The wind produces a horizontal drag $F_d = 1.20\ \text{N}$ opposing forward motion. Find the net horizontal acceleration.

Solution

As before, required total thrust $T = \dfrac{mg}{\cos\theta}$.

Compute $mg = 2.00 \times 9.81 = 19.62\ \text{N}$. $\cos 15^\circ \approx 0.9659258$.

So $T \approx 19.62/0.9659258 \approx 20.3121\ \text{N}$.

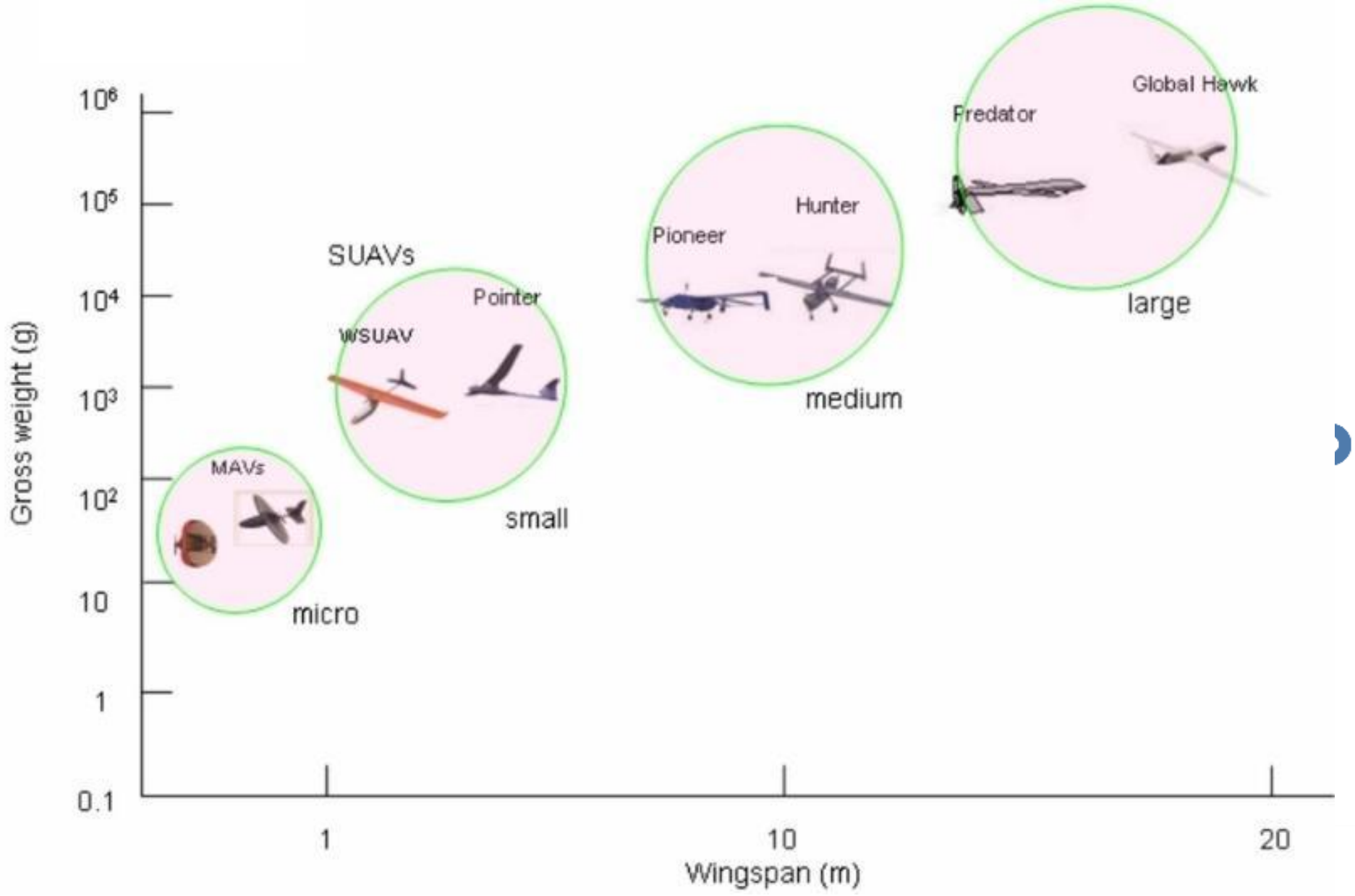
Horizontal thrust component: $F_h = T \sin 15^\circ$. $\sin 15^\circ \approx 0.2588190$.

$F_h \approx 20.3121 \times 0.2588190 \approx 5.2572\ \text{N}$.

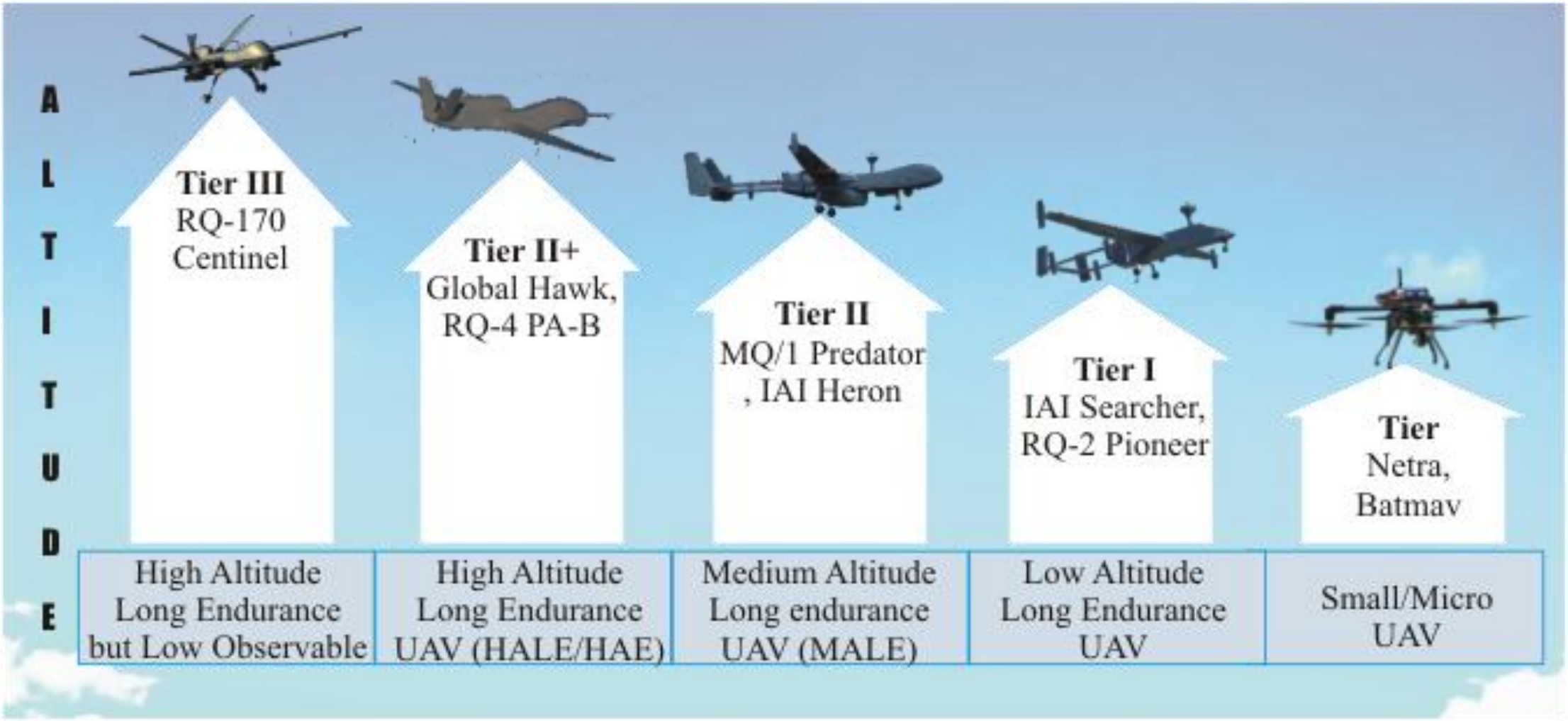
Net horizontal force $F_{net} = F_h - F_d = 5.2572 - 1.20 = 4.0572\ \text{N}$.

Horizontal acceleration $a_x = F_{net}/m = 4.0572/2.00 = 2.0286\ \text{m/s}^2$.

Weight based classification



Altitude based classification



Quadcopter



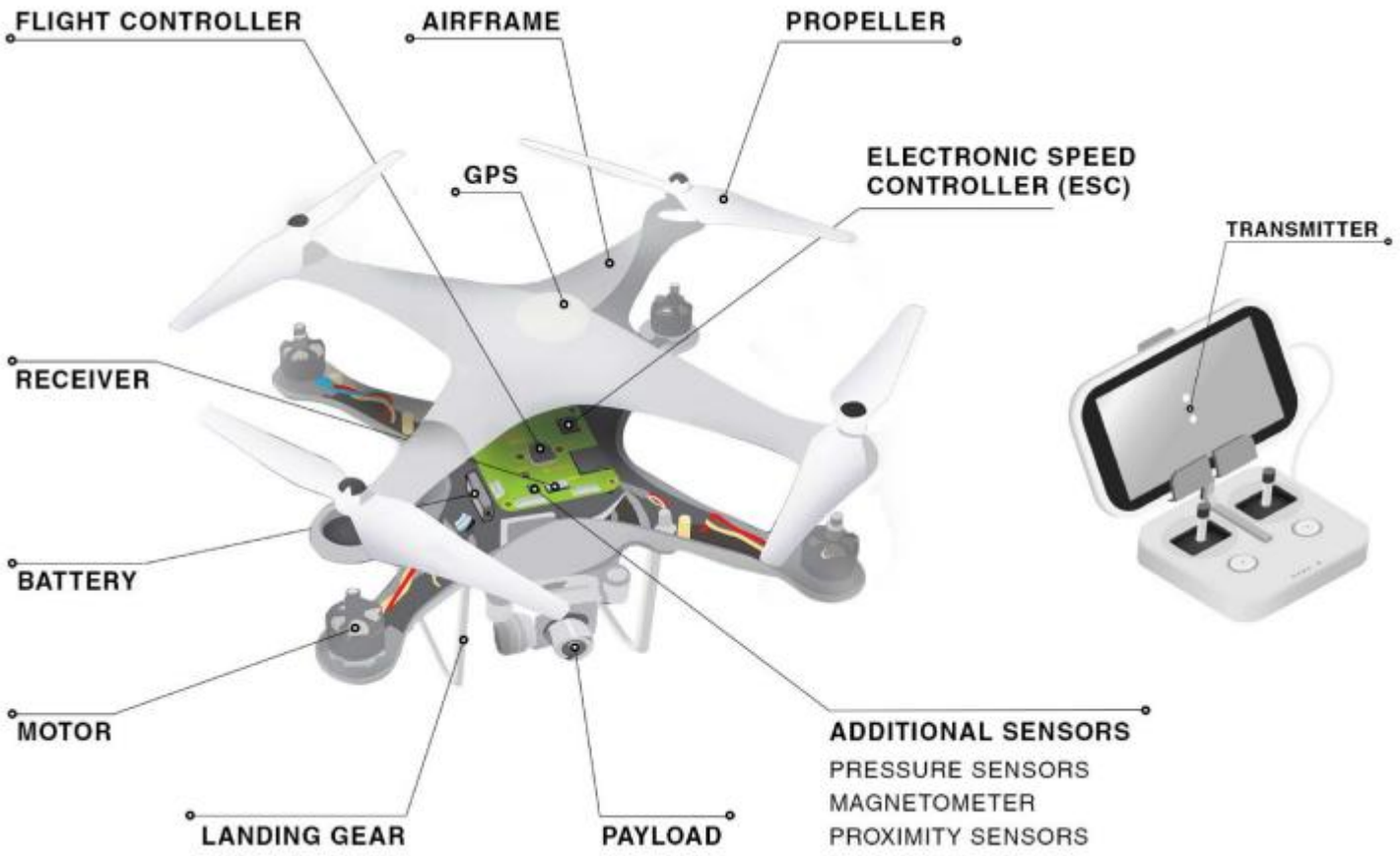
Hexacopter and Octacopter



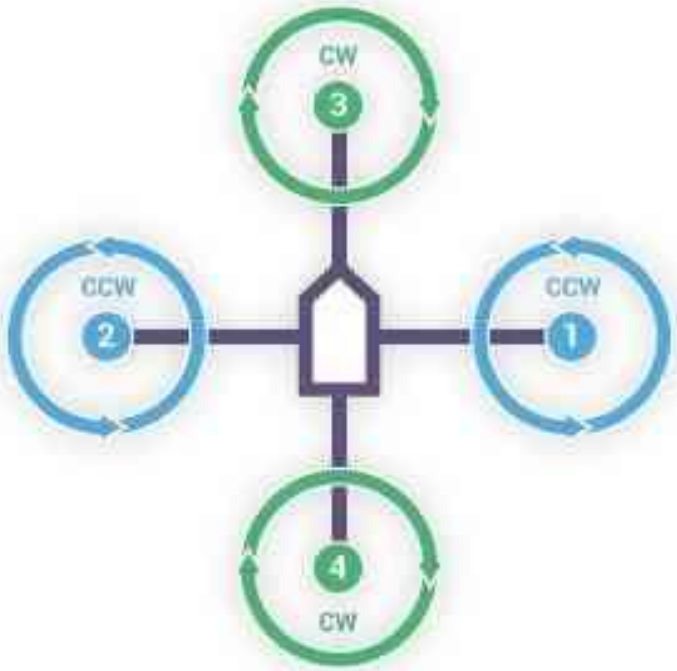


Drone Components

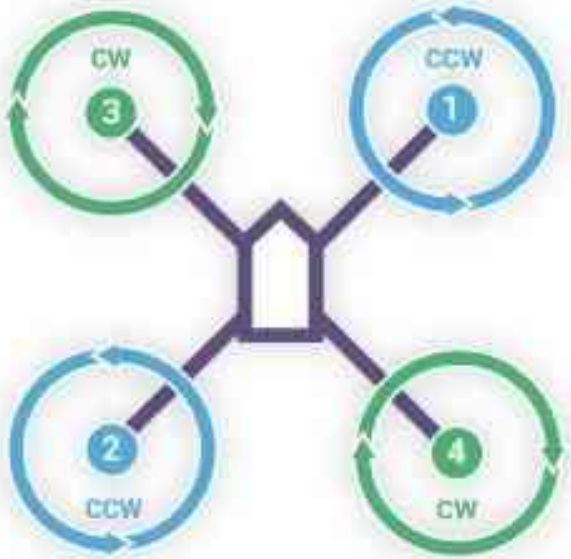
- Frame
- Motor
- Electronic speed controller
- Flight Controller
- Propellers
- Batteries – Lithium Polymer



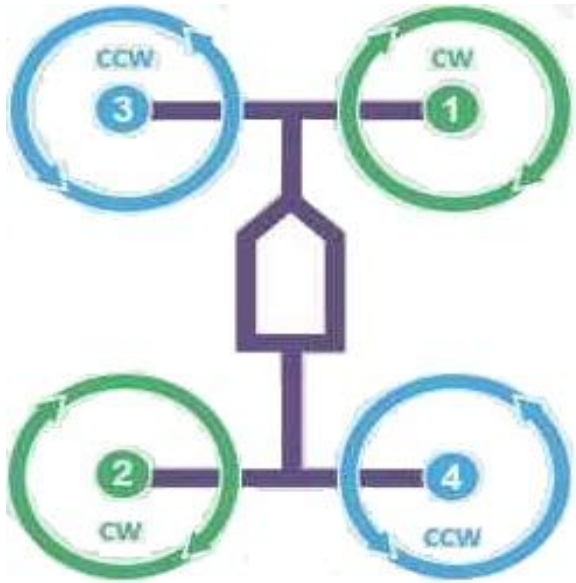
Frame



QUAD +



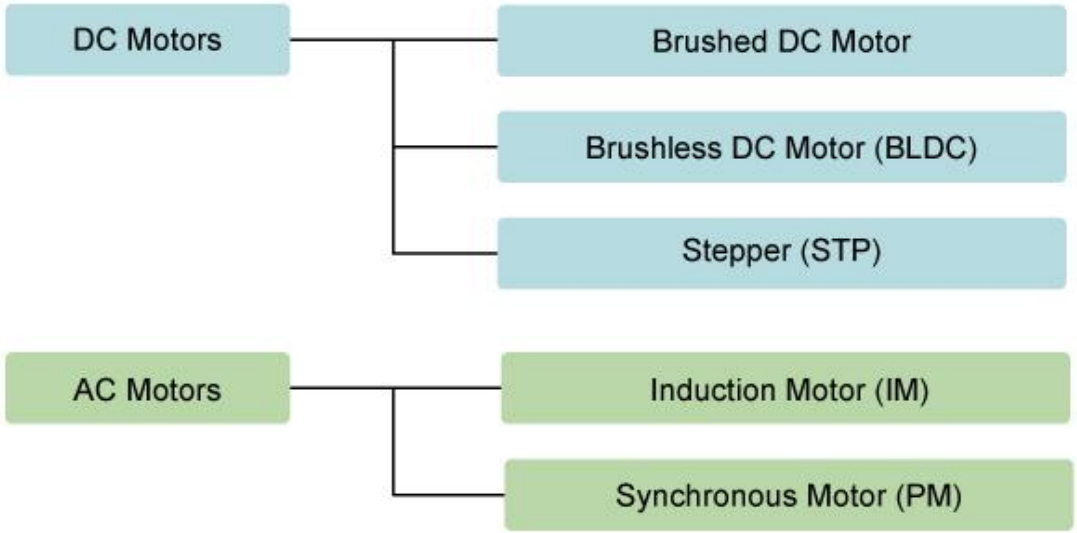
QUAD X



QUAD H

Motors

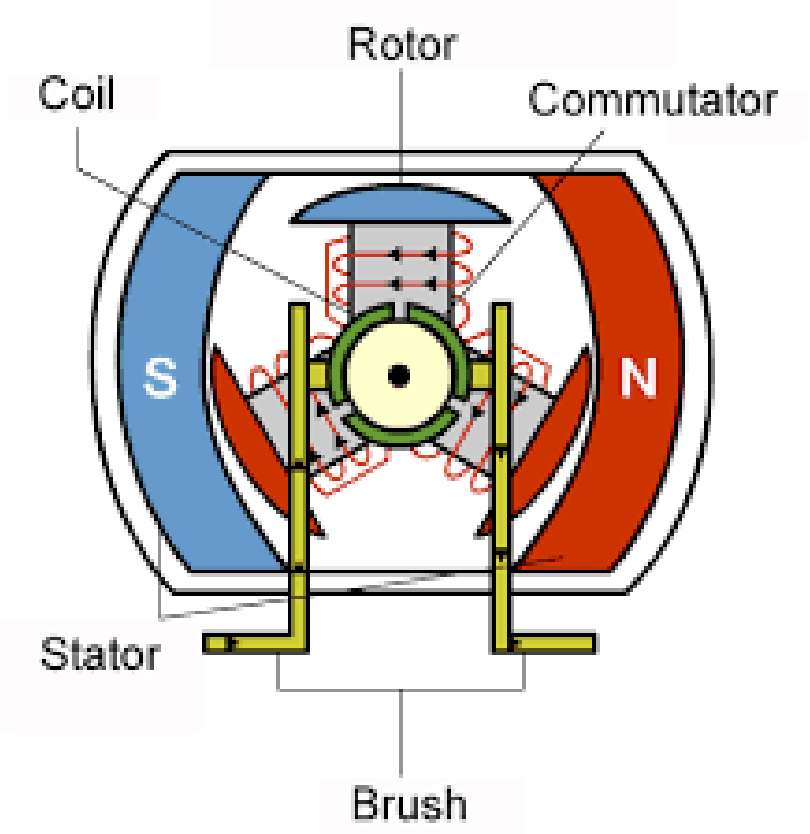
- **Common Motor Types**



Brushed DC Motors

- simple design and easy control
- widely used to open and close disk trays. In cars, they are often used for retracting, extending, and positioning electrically-powered side windows.





Brushless DC Motors

- A brushless DC electric motor (BLDC motor or BL motor), also known as an electronically commutated motor (ECM or EC motor) or synchronous DC motor, is a synchronous motor using a direct current (DC) electric power supply.

- It uses an electronic controller to switch DC currents to the motor windings producing magnetic fields which effectively rotate in space and which the permanent magnet rotor follows.

- The advantages of a brushless motor over brushed motors are high power-to-weight ratio, high speed, nearly instantaneous control of speed (rpm) and torque, high efficiency, and low maintenance

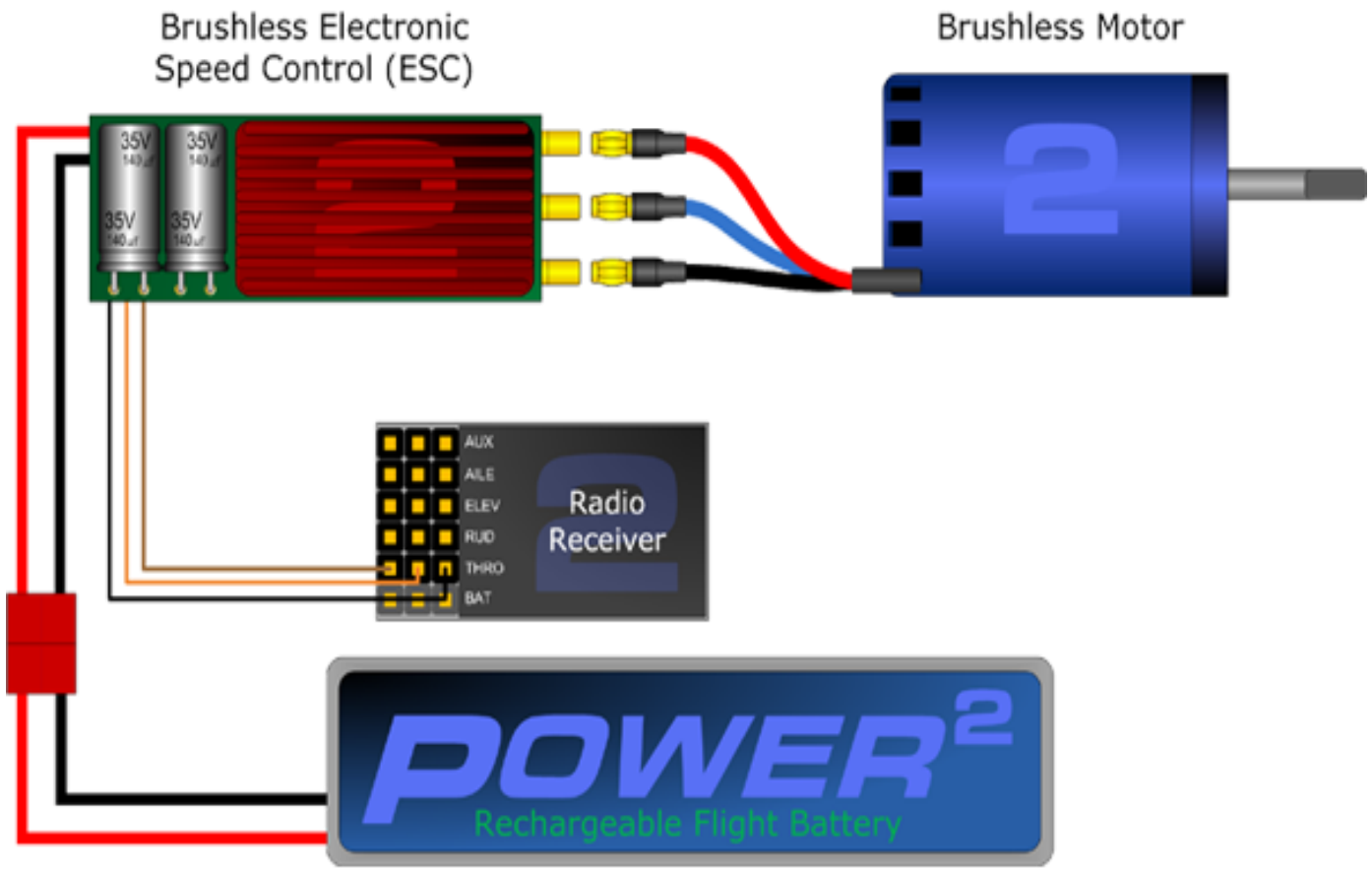
Brushless DC Motors

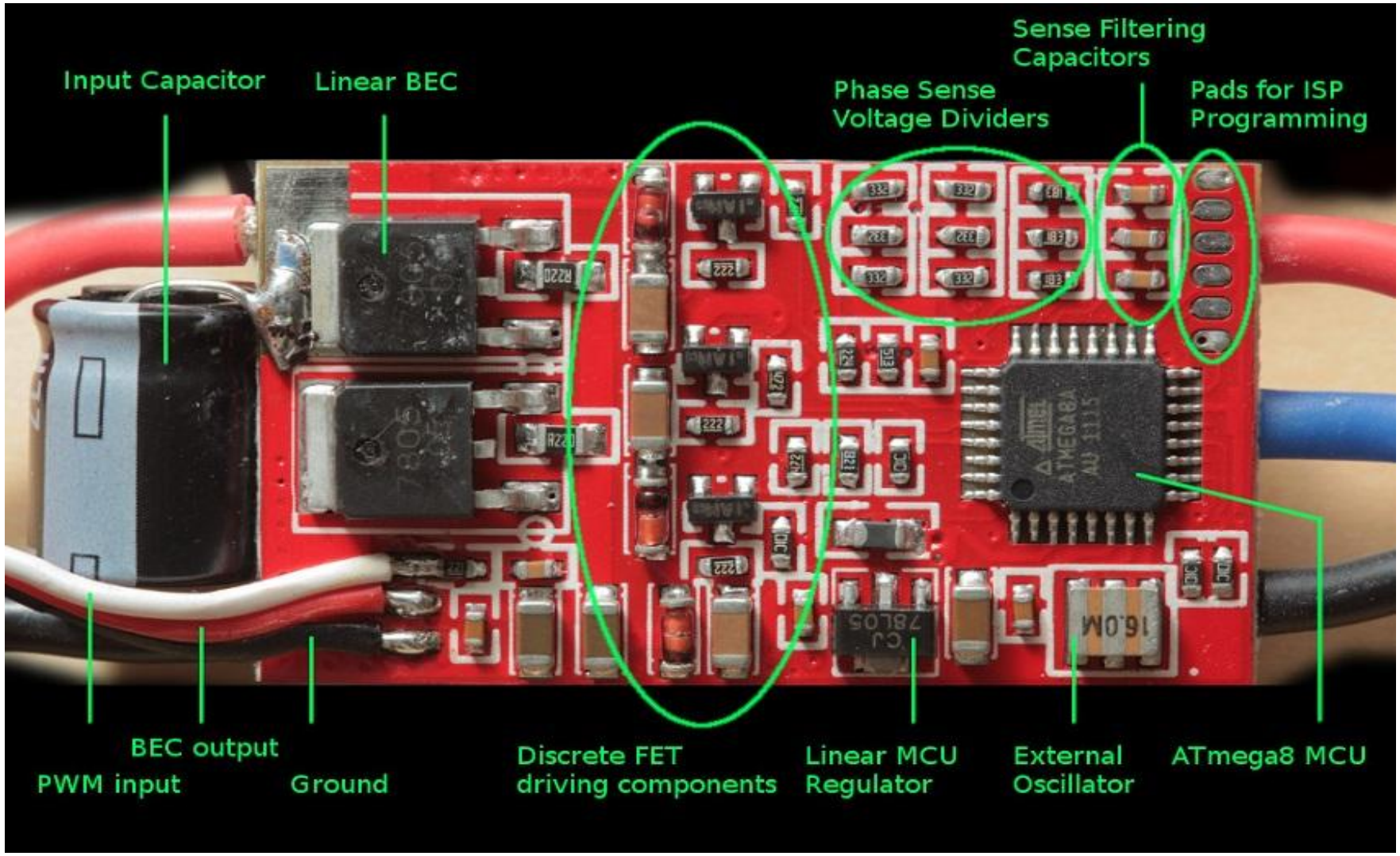


Electronic Speed Controller

- An electronic speed controller (ESC) is an electronic circuit used to vary the speed and direction of an electric motor.
- **Linear BEC's**:- A linear BEC basically uses a resistor to step the voltage down from the 7.4v-11.1v we commonly use with our Lipo packs down to the 5 volts required for the receiver.
- **Switched BEC’s:** Switched BEC's actually switch the voltage on and off very rapidly to effectively drop the voltage without generating heat or wasting electricity.

Brushless Motor Wiring

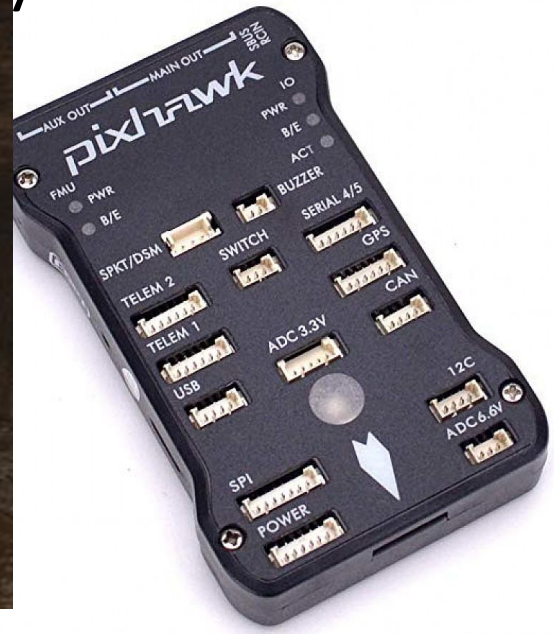



Input Capacitor
Linear BEC
Sense Filtering Capacitors
Phase Sense Voltage Dividers
Pads for ISP Programming
PWM input
BEC output
Ground
Discrete FET driving components
Linear MCU Regulator
External Oscillator
ATmega8 MCU

Flight Controller

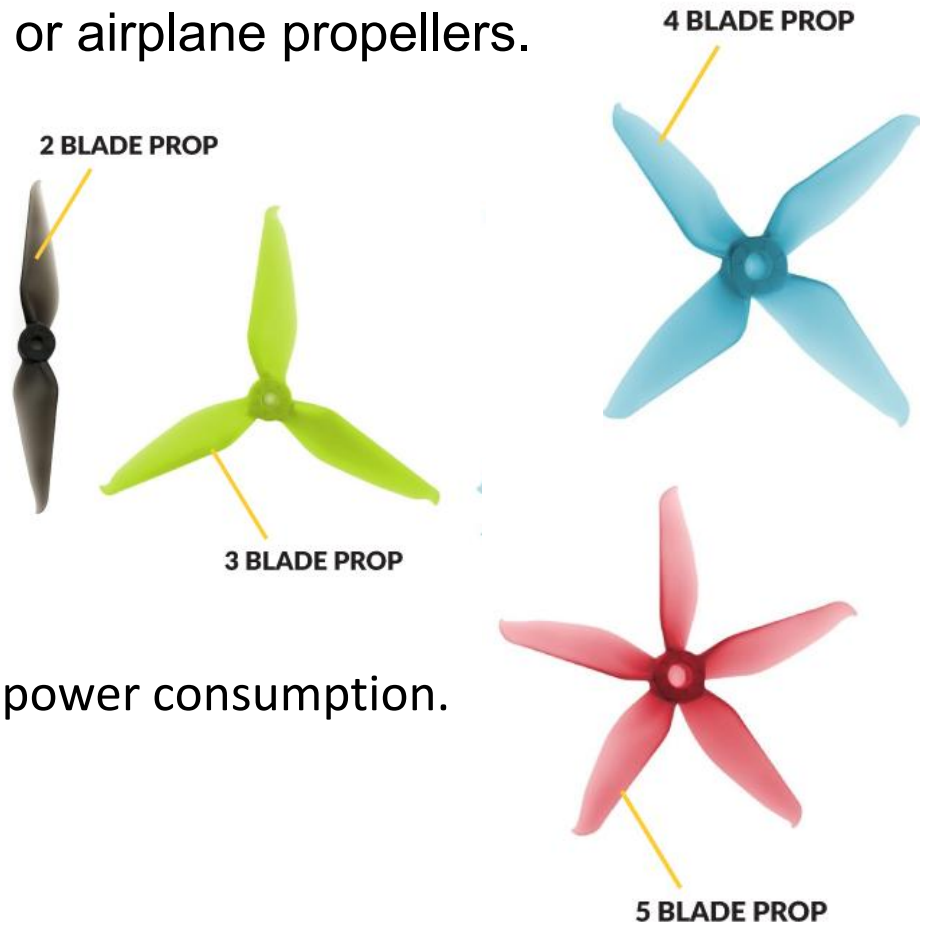
- The flight cor functioning of the drone.
- Directional co y load control.
- It is basically
- Pixhawk, DJI





Propellers

•Definition: A propeller is an aerodynamic device that converts rotational motion into thrust.
•Importance: Determines **lift, stability, maneuverability, and efficiency** of the drone.
•Real-life analogy: Compare with fan blades, helicopter rotors, or airplane propellers.



Types of Drone Propellers

•Based on Material:

- Plastic → lightweight, cheaper, but less durable.
- Carbon Fiber → strong, durable, high efficiency, expensive

•Based on Blade Count:

- 2-blade → efficient, higher speed, longer flight time.
- 3-blade or 4-blade → more thrust, better stability, but more power consumption.

Propellers

Physics of Propellers

- **Thrust Generation:** Explained via **Newton's Third Law** (propeller pushes air down, drone goes up).
- **Lift Equation (simplified):**

$$T \approx C_T \cdot \rho \cdot n^2 \cdot D^4$$

- where:
  - T= Thrust
  - C_T =Thrust coefficient (depends on shape)
  - ρ= Air density
  - n= Rotations per second
  - D= Propeller diameter

Propellers

Propeller Nomenclature & Dimensions

•Propeller size is written as **Diameter × Pitch** (e.g., 10 × 4.5).

- **Diameter (10 in)** → distance tip-to-tip; affects thrust & stability.
- **Pitch (4.5 in)** → distance a propeller would move forward in one rotation (like a screw); affects speed.

Clockwise (CW) vs Counterclockwise (CCW)

• Drones need opposite propeller directions to **cancel torque**.

• One motor with propeller spins the whole body → why opposite rotation is needed.

Propellers

Applications & Design Considerations

- Racing drones → small, high-pitch propellers (speed).
- Camera drones → larger, slow-spinning props (stability & efficiency).
- Heavy-lift drones → multi-blade, wide props (high thrust).

Propeller Pitch

- **Definition:** The distance a propeller would theoretically move forward in one complete rotation if it were moving through a solid medium (like a screw through wood).
- **Unit:** Usually in **inches** (e.g., 4.5 in).
- **Effect:**
  - **High pitch** → pushes more air per rotation → higher speed, more load on motor.
  - **Low pitch** → less air per rotation → slower but more efficient and stable.
- **Analogy:** Like bicycle gears: low pitch = easy pedaling, high pitch = harder pedaling but faster speed.

Propellers

Propeller Span (Diameter)

- **Definition:** The total distance from one tip of the propeller blade to the opposite tip (like wingspan).
- **Unit:** Inches or centimeters.
- **Effect:**
  - **Large span (big propeller):** More thrust, better efficiency, but slower response.
  - **Small span (small propeller):** Less thrust, quicker response, good for racing drones.
- **Analogy:** Like the size of a fan — bigger fans move more air, smaller fans are quicker but weaker.

Quick Example: 10×4.5 Propeller

- **10** → Span/Diameter = 10 inches
- **4.5** → Pitch = 4.5 inches (forward distance per rotation)

So, a **10×4.5 propeller** is large (good thrust & stability) with medium pitch (balanced speed & efficiency).

Propellers

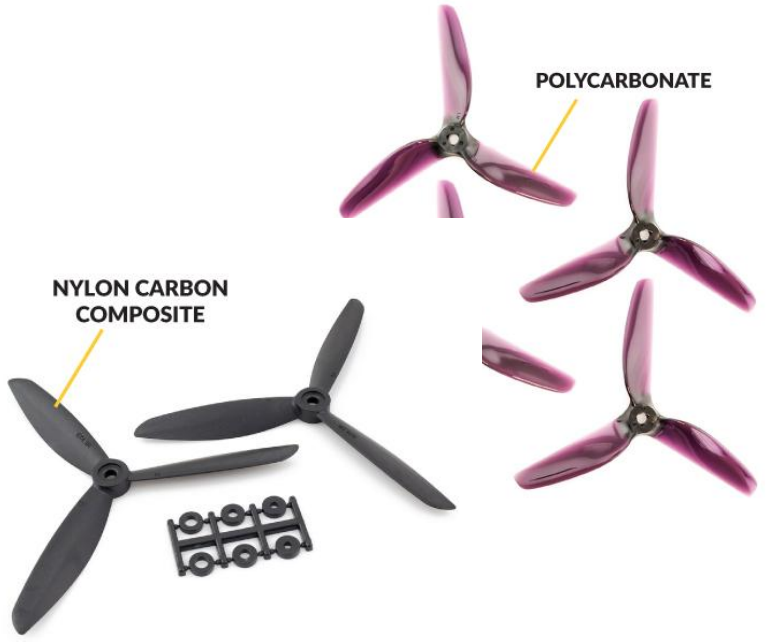
Key Properties of Drone Propellers

1. Blade Count

- **2-blade:** Most efficient, longer flight time, but less thrust.
- **3-blade:** Balance of thrust & efficiency, common in racing drones.
- **4 or more blades:** More thrust, more stability, but higher power consumption.

2. Blade Shape (Airfoil Profile)

- Narrow blades → efficient, less drag, better for endurance.
- Wide blades → push more air, better for lifting heavy loads.
- Curvature of airfoil affects **lift coefficient** and noise.



3. Material

- **Plastic:** Cheap, flexible, absorbs crashes, less efficient.
- **Carbon Fiber:** Strong, rigid, efficient, expensive.
- **Wood (rare in drones):** Stable, low vibration, but not durable.

Propellers

4. Weight & Inertia

- Heavier propellers need more torque to spin up → slower motor response.
- Lighter propellers give **faster acceleration and maneuverability**.

5. Stiffness / Flexibility

- Rigid props (carbon fiber) → consistent thrust, less deformation, efficient.
- Flexible props (plastic) → bend under load, absorb shock, safer for beginners.

6. Rotation Direction

- **CW (clockwise)** and **CCW (counterclockwise)** props are paired to balance torque.
- Wrongly installed props → drone flips instantly.

Propellers

7. Hub Size & Mounting Style

- Bore diameter (hole size in the center).
- Mounting types: press-fit, screw-on, or quick-release.

8. Noise Characteristics

- Larger, low-pitch props = quieter, lower RPM.
- Smaller, high-pitch props = noisier, higher RPM.

9. Efficiency

- Combination of **thrust produced per watt of power consumed**.
- Affected by pitch, diameter, material, and blade shape.

Propellers

A drone propeller has:

- Diameter = **10 in (0.254 m)**
- Pitch = **4.5 in (0.114 m)**
- Rotating speed = **6000 RPM (100 rev/sec)**
- Air density = **1.225 kg/m³**
- Thrust coefficient $C_T = 0.1$

Question: Calculate the approximate thrust produced by the propeller.

Solution:

Formula:

$$T = C_T \cdot \rho \cdot n^2 \cdot D^4$$

where

- $n = 100\ \text{rev/s}$
- $D = 0.254\ \text{m}$

$$T = 0.1 \times 1.225 \times \Big(100\Big)^2 \times \Big(0.254\Big)^4$$

$$T \approx 5.0\ \text{N}$$

So, the propeller produces **~5 N thrust** at 6000 RPM.

Payloads

- Payload refers to the additional weight a drone can carry.
- Critical for determining the drone's application and capabilities.
- Impacts flight time, stability, and operational efficiency.

- Cameras
- Sensors
- Delivery systems
- LIDAR systems
- Communication Equipment
- Weapon systems
- Other miscellaneous.

Payloads…

Cameras

- RGB cameras
- Multispectral
- Thermal

- Aerial photography and videography
- Inspection and surveillance
- Agriculture monitoring
- Defence

Sensors

- Optical sensors
- Infrared sensors
- Chemical sensors
- Others

- Environmental Monitoring
- Industrial inspections
- Scientific research

Payloads…

Delivery systems

- Drop systems
- Winch systems
- Holding systems

- Package delivery
- Medical supply transport
- Disaster relief

LIDAR systems

- Light detection and ranging
- Sense detailed 3D information

- Topographic surveys
- Forestry management
- Urban vegetation
- Infrastructure inspections

Payloads…

Communication Equipment

- Relay stations
- Sound systems
- Signal boosters

- Communication in remote areas
- Extended control range

Weapon systems

- Missiles
- Bombs
- Loaded poison or chemicals

- Military operations
- Combat support
- Enemy movement detection
- Weapon dumps

Payload controls

Factors considered

- Weight
- Power consumption
- Cost
- Compatibility

Challenges

- Payload capacity
- Stability
- Battery life
- Integration modules

Payload Integration

- Compatibility
- Weight balancing
- Secure Mounting
- Test and Calibrate
- Intra-system communication

Factors effecting Payload Capacity

- **Propeller Size & Efficiency** – larger props = more lift.
- **Motor Power (kV rating)** – stronger motors can spin faster or handle larger props.
- **Battery Capacity** – more payload = more current draw = reduced flight time.
- **Frame Strength** – structure must hold the extra weight.
- **Number of Rotors** – quad (4), hexa (6), octa (8) → more rotors = more thrust available.

Rule of Thumb

Each motor should produce at least **2× the thrust required for its share of the weight** (safety margin).

Payload Capacity

A quadcopter has four motors, each producing a maximum thrust of **8 N**.
The drone itself (frame + battery + electronics) weighs **2 kg**.
Question: What is the maximum payload it can carry?

Solution:

- Total thrust available = $4 \times 8 = 32\ \mathrm{N}$
- Drone weight = $2 \times 9.81 = 19.6\ \mathrm{N}$
- Maximum lift for payload = $32 - 19.6 = 12.4\ \mathrm{N}$
- Payload capacity = $\frac{12.4}{9.81} \approx 1.26\ \mathrm{kg}$

Answer: The drone can carry about **1.3 kg** safely.

Future Trends

Advancements

- Miniaturization
- Increased efficiency
- AI integration
- Autonomous payload management
- Multiple payload capacity
- Strategic integration